Price Matching and Edgeworth Cycles*

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Abstract

With price matching policies, retailers offer consumers to match prices offered by cheaper competitors. This article investigates how price matching affects pricing decisions in dynamic markets with cycling prices. The model predicts price-matching retailers to post higher prices and to lead price restorations. The lower bound of price undercutting increases for price-consulted retailers. Consulted stations also anticipate earlier price restoration reactions from implementing stations and, thus, provoke restorations by executing more price decreases earlier. The last effect dominates in welfare calculations, such that price matching has positive welfare implications. The predictions are verified with price data on the German gasoline retail market. Making use of the discontinuity of two policies, I disentangle the competitive effects on implementing and price-consulted market participants. In line with the model, a price-matching retailer increases his promotional prices significantly with the introduction of price matching.

Keywords: Low-price guarantees; Dynamic pricing; Retail gasoline

JEL classification: L11; L71; L81

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1 Introduction

During recent years, price-matching policies regained popularity with the adoption by various U.S. retailers like Wal-Mart, Target and Best Buy. With price matching, retailers offer consumers the ability to match lower competitor's prices. Contrary to the intuition that price matching is an indicator of high competition and a handy device for consumers to shop for the cheapest price of their most preferred retailer, literature in economics dominantly predicts negative welfare effects (Hay, 1982; Salop, 1986). Lower competition, higher opportunities to collude, and a higher degree of price discrimination (Corts, 1997) arising from price-matching policies that are mentioned as objectives for implementation. Empirical studies are scarce and appropriate data sets to distinguish price interactions are rarely available. Therefore, while economic theory would predict negative welfare implications from price matching, newspapers equate price matching with price wars and rate them as a competition-accelerating device (Hviid and Shaffer, 1999).

The existing economic literature on price matching has predominantly focused on static models of interaction among retailers. However, real-world markets are increasingly dynamic. This is evident, inter alia, in the many retail gasoline markets featuring price cycles (Noel, 2007; Isakower and Wang, 2014). The Edgeworth Cycle model by Maskin and Tirole (1988) features dynamic interactions and rationalizes price cycles in a competitive market environment. This article analyzes the implications of price matching in such a dynamic market setting, where dynamic interactions are essential to explain the firms' strategies. The dynamic component complicates to solve the model by two factors. i) It is challenging to determine an analytic solution of the model and ii) the equilibrium selection is ambiguous. By using the homotopy analysis method by Eibelshäuser and Poensgen (2019b), this article overcomes both challenges and identifies the limiting quantal response equilibrium, which turns out to select an equilibrium that coincides with the empirics.

My model shows that the implementation of price matching reduces firms' intertemporal price spread. A policy-implementing firm (denoted as implementing firm, hereafter) generally posts higher prices with price matching. The necessary level that provokes price restoration is situated at a higher price leading to earlier price provocations by the price-consulted (i.e. the firm of which prices are referenced; denoted as consulted, hereafter) firm. Thus, the implementing firm initiates more price restorations. Consumers, in turn, shop more frequently at medium-low prices because consulted firms conduct stronger price reductions.

To test the theoretical predictions, I exploit the responses to price-matching policies in the

German retail gasoline market. Using a data set containing price information of the entire market on a minute level, this article analyzes the price effects for implementing consulted retailers. Implementing and consulted stations post higher prices after the introduction of price matching by a large retailer. The reactions to a policy of a relatively small retailer do not fully comply with the model's predictions.

The theoretical literature on price matching is divided into three static model types that describe how such a policy may affect pricing strategies of firms. In contrast to the predictions of this article's dynamic model, static models overwhelmingly predict negative welfare effects of price-matching as it leads to supra-competitive pricing (Corts, 1997). A summary on the predictions is provided in table 1.

Hay (1982) and Salop (1986) investigate the implications of price matching on competition. Fewer customers are attracted from price reductions as they can claim that same price at a price-matching firm and price-matching policies reduce incentives for consulted firms to undercut price-matching competitors. Hence, consulted firms remain to undercut competitors and the intensity of competition decreases.

Similarly, Arbatskaya (2001) claims that price matching can deter entry. In situations where market entry requires a significant price discount to attract attention and to stimulate the customers to switch to the new entrant, price matching lowers attention for the price-matched entrant. Therefore, price matching can be a threat for new market entrants, and a useful device for oligopolists that face low competition.

Furthermore, price matching can stabilize cartels. First, the policy itself might serve as a credible punishment device to deter cartel members from deviation (Lu and Wright, 2010). Even though it might be inefficient to decrease the cartel price ex-post when small cartel members deviate, price matching is a hard commitment for the cartel participants to react to deviators. Therefore, a potential deviator anticipates that competition will be intense after deviating, and has a lower incentive to end the cooperation. Furthermore, a price-matching policy is able to outsource costs of monitoring deviants to customers (Hviid, 2010). Customers alert cartel members of a deviant when claiming price matches. Hence, monitoring may be less costly and, therefore, more efficient than monitoring colluding firms directly.

Hviid and Shaffer (1999) show that effects of reducing competition decrease significantly when hassle costs are introduced. Hassle costs occur for customers that claim price-matching guarantees[1]. They show that the effect of hindering competition is negligibly small when hassle

[1]Examples of hassle costs are time and effort necessary to request refunds, to discuss and prove that a cheaper

Table 1: Summary of Price Matching Models on their Predictions on Posted Prices

Model	*Reference*	*Implementing Firms*	*Consulted Firms*
Cartel-facilitating device	Hay (1982)	+	+
Price-discrimination device	Corts (1997)	+/-	+/-
Low-price signal	Manez (2006)	+	-

costs exist. In addition, Hviid and Shaffer (1999) argue that many price-matching implementing retailers introduce high hassle costs by implementing difficult terms and conditions, thus restricting the de facto use of price-matching policies.

Corts (1997) shows that price matching is a possible price discrimination device[2]. Firms try to attract price-sensitive (sophisticated) customers who will always shop for the lowest price by maintaining the ability to skim off profits from less price-sensitive (unsophisticated) customers with high prices. Generally, price-sensitive customers are better off, while unsophisticated customers can be disadvantaged by the policy. Depending on the model's parameters, prices may either rise or decrease for all firms. With high transportation costs, and, consequently, sophisticated customers having a low demand elasticity, the introduction leads to more intense competition between retailers and reduced posted prices. Low transportation costs make the sophisticated consumers' demand more elastic and have exactly the opposite effect.

Moorthy and Zhang (2006) develop a model in which price-matching policies signal low prices in equilibrium with three assumptions: i) substantial cost differences between the firms, ii) sufficient number of customers who activate the guarantee and iii) customers that still shop at the high-cost (and thus high-priced) firms even though low prices are successfully signaled. Compared to the equilibrium without price matching, low-cost firms increase prices because their credible signal attracts customers who are not aware of the competitors' prices. High-cost firms have to decrease their prices in comparison to the equilibrium where price matching is not allowed. Thus, welfare effects are not clear-cut. Manez (2006) provides a numerical example with reasonable parameter values in which customers are better off after the implementation of a price-matching policy.

Hess and Gerstner (1991) who analyze supermarket chains that promised to match prices of competitors conducted the first empirical investigations concerning price matching. An implementing chain published a so-called, "Price Finder" that reported its own prices and those of some of its competitors every week. Hess and Gerstner (1991) use these published prices to

competitor exists, and that everything is in line with the terms and conditions of the price-matching policy.

[2]See Stole (2007) for an extensive review on price discrimination.

compute a basket-related relative price for every retailer. They find supermarkets to converge in their prices and to increase prices after introduction. Therefore, they take the pricing responses as evidence that price matching supports price coordination.

Arbatskaya (2001), Arbatskaya et al. (2004) and Arbatskaya et al. (2006) find weak support for price matching being anti-competitive and therefore, leads to higher prices. Arbatskaya et al. (2004) investigate the patterns and restrictions of price-matching policies. The observed restrictions tend to apply only to rival firms' advertised prices or they are associated with high hassle costs. Therefore, the restrictions do not appear to be consistent with the theoretical models of price matching.

In a to this article independent research project, Dewenter and Schwalbe (2016) analyze a similar data set on the German retail gasoline market. Using a difference in difference estimator and referring their results to static models, they conclude that price matching has competition reducing effects. My article shows that the integration of dynamic interactions reveals another mechanism that is beyond those explored by traditional static models.

In contrast to static models, the dynamic model in this article indicates that price matching has positive welfare implications in markets with Edgeworth Cycles. The model is more robust to capture the price implications than existing models because it integrates the strategic component of dynamic interactions that drive price cycles. My results challenge the conventional wisdom that price matching unavoidably leads to competition reduction and suggests to reconsider antitrust actions against price-matching policies.

The remainder of the article is structured as follows: The next section derives an Edgeworth Cycle model that integrates price matching to determine testable hypotheses. The third section introduces the data set, explains the market setting, and tests the hypotheses. The fourth section compares the empirical findings with static price-matching models. The final section concludes.

2 Dynamic Model

Cycling prices are observed in markets with substitutable products and market transparency, in particular online search auctions (Zhang and Feng, 2011) and retail gasoline markets (Noel, 2007). Eibelshäuser and Wilhelm (2019) show that an extended Edgeworth Cycle model is able to explain the incremental implementation of additional price cycles in the German market

between 2014 and 2018. Static models are not able to generate the saw-tooth shaped price pattern observed in different markets with a low level of product differentiation and price transparency.

The model is based on the work by Maskin and Tirole (1988) and the chosen parameters relate to their example. Two firms ($i = 1, 2$) compete with a homogeneous product with zero costs in discrete time with an infinite horizon. Total demand[3] is given by $D(p) = 2 - \min(p_t^1, p_t^2)$ and profits are given by

$$\pi^i(p_t^1, p_t^2) = \begin{cases} \pi(p_t^i) & \text{if } p_t^i < p_t^j \\ \pi(p_t^i)/2 & \text{if } p_t^i = p_t^j \\ 0 & \text{if } p_t^i > p_t^j. \end{cases}$$

Firms discount the future with the discount factor $\delta = 0.95$. Firms alternately choose their posted prices. Firm 1 posts price $\hat{p}_t^1$ in odd-numbered periods and firm 2 price $\hat{p}_t^2$ in even-numbered periods. Firms post prices such that their discounted future profits are maximized.

In the traditional version of this model, effective prices coincide with posted prices. With price matching, the effective price of firm 1 is the lowest available price at the time point t,

$$p_t^1 = \min(\hat{p}_t^1, \hat{p}_t^2)$$

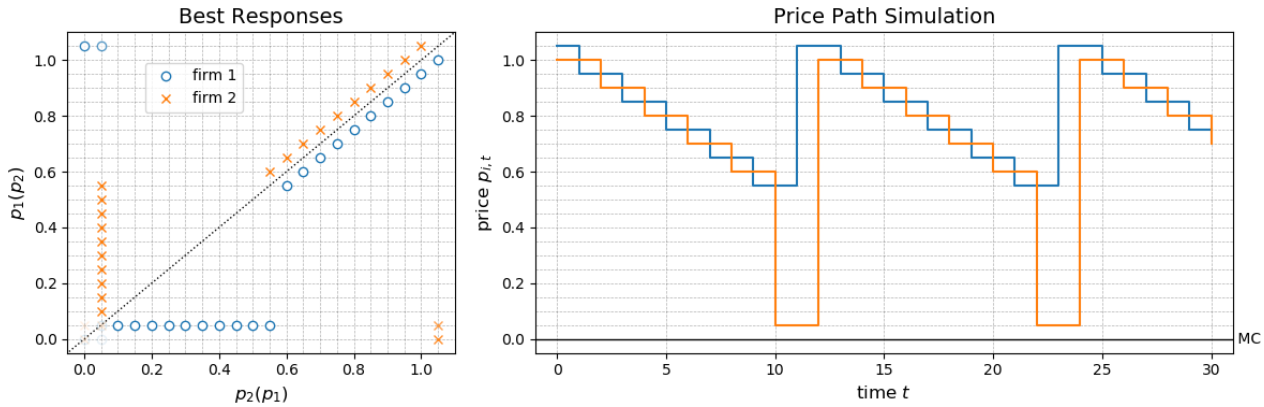
while the effective price for firm 2, which does not introduce price matching is always the posted price $p_t^2 = \hat{p}_t^2$. As effective prices represent transaction prices, they determine profits for all firms.

As a last extension, only a part of consumers reacts on price matching. In this case, only a share of $0 \leq \alpha \leq 1$ (redemption rate) consumers are willing or able to claim price matches. In this extension, effective prices attract only a proportion of α consumers, while the remaining share of $1 - \alpha$ customer react to posted prices only. Integrating partial price matching and solving for posted prices gives a profit function of

$$\pi^1(\hat{p}_t^1, \hat{p}_t^2) = \begin{cases} \pi(\hat{p}_t^1) & \text{if } \hat{p}_t^1 < \hat{p}_t^2 \\ \pi(\hat{p}_t^1)/2 & \text{if } \hat{p}_t^1 = \hat{p}_t^2 \\ \alpha\pi(\hat{p}_t^2)/2 & \text{if } \hat{p}_t^1 > \hat{p}_t^2 \end{cases}$$

[3] All results are robust to different parameterized linear demand functions.

Figure 1: Edgeworth Cycle Equilibrium Without Price Matching



The left graph represents the best responses strategy space for both players conditional on the action of the other player. The right graph draws a simulated price path of 30 periods within the infinite game.

for the price matching implementing firm and a the profit function of

$$\pi^2(\hat{p}_t^1, \hat{p}_t^2) = \begin{cases} (1 - \frac{\alpha}{2})\pi(\hat{p}_t^2) & \text{if } \hat{p}_t^2 < \hat{p}_t^1 \\ \pi(\hat{p}_t^2)/2 & \text{if } \hat{p}_t^2 = \hat{p}_t^1 \\ 0 & \text{if } \hat{p}_t^2 > \hat{p}_t^1. \end{cases}$$

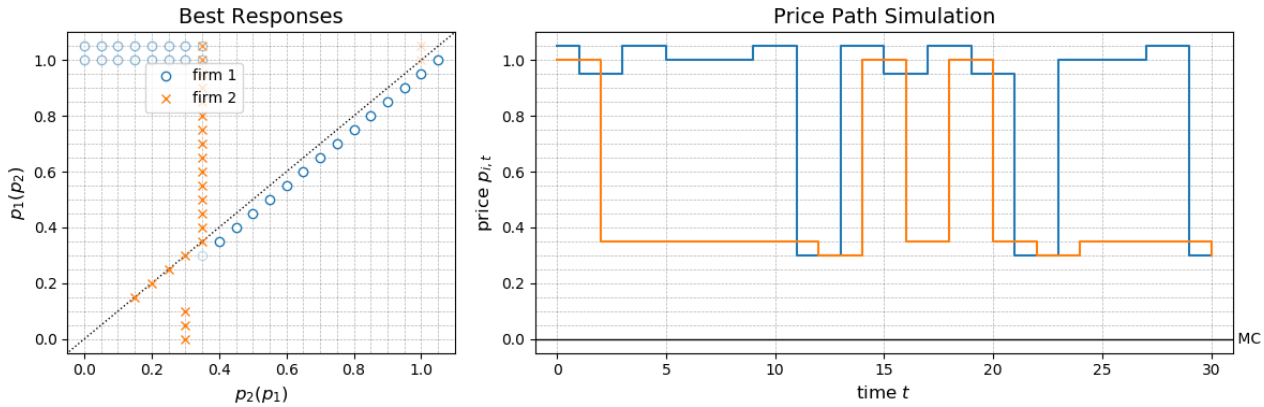
for the price-consulted firm.

Maskin and Tirole (1988) show that cycling prices constitute a Markov perfect equilibrium. To select an explicit equilibrium, this article uses the limiting Markov quantal response equilibrium (QRE). This concept is introduced by McKelvey and Palfrey (1995, 1998) for normal-form and extensive-form games, respectively. The concept of logit Markov QRE and a solution algorithm are first introduced by Breitmoser et al. (2010), and formally defined by Eibelshäuser and Poensgen (2019b). In the quantal response framework, players are assumed to play optimal strategies only with some errors. As the error rate tends to zero, a Markov perfect equilibrium is obtained, the so-called limiting Markov quantal response equilibrium. I use the software of Eibelshäuser and Poensgen (2019a) to determine the equilibrium computationally.

Figure 1 illustrates the equilibrium for the classic Edgeworth Cycle model with a discrete price strategy grid, $p_t^i, \hat{p}_t^i \in \{0, 0.05, 0.10, ..., 1.05\}$, representing all possible prices that the players can post.[4] The left part of the figure represents the best responses of both players. The right

[4]Prices above 1.05 are excluded without loss of generality. The competitor's strategic response to a prices above the monopoly price is a price that is smaller or equal the monopoly price. Therefore, firms do not have any advantage to play prices that are higher than one tick size above the monopoly price.

Figure 2: Edgeworth Cycle Equilibrium With Price Matching



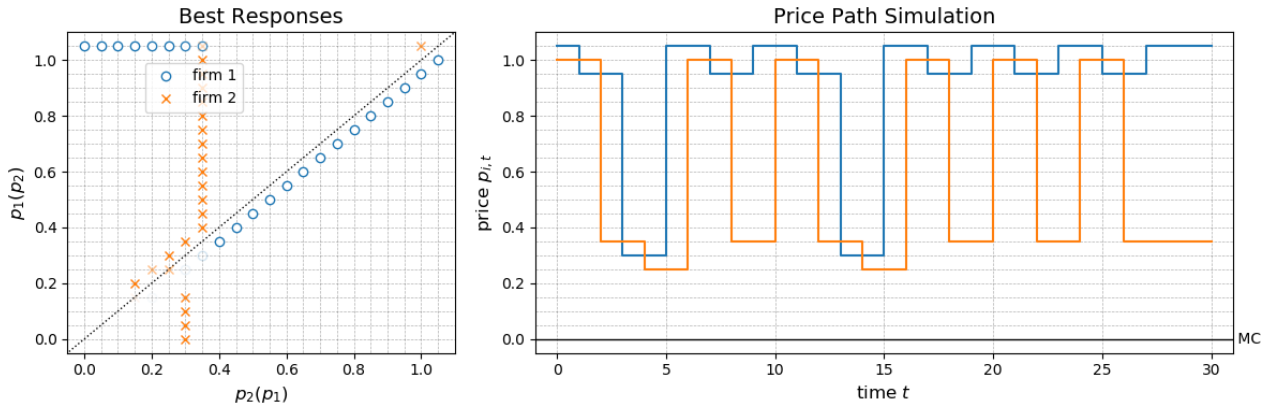
The left graph represents the best responses strategy space for both players conditional on the action of the other player. The right graph draws a simulated price path of 30 periods within the infinite game.

part plots a simulated price path. The equilibrium consists of an undercutting phase where firms alternately absorb competitors demand by slightly undercutting each other. Once firms reach a level of around half the monopoly price, one player sets a price close to the marginal cost to provoke a price restoration. The other firm plays a mixed strategy at this point, and either restores the price or also reduces the price to marginal cost, to provoke the other firm to restore the price. This phase represents the so-called war of attrition. Once a firm restores the price, the other firm slightly undercuts this price and the undercutting phase restarts.

Figure 2 visualizes the model with implemented price matching by firm 1. With price matching, the undercutting phase almost disappears. While both players slightly undercut each other at specific points, the price-matching player initiates price restorations at a much higher price level as he benefits from undercutting his competitor and losses from being undercut are reduced. The consulted player anticipates that the implementing firm is willing to restore prices earlier and utilizes this behavior by provoking price restorations earlier in the price undercutting phase of the game.

Figure 3 plots the equilibrium with partial price matching when 30 % of customers are willing to claim lower prices. Again, the policy leads to price restorations at a higher price level. The model predicts higher probabilities for players to undercut each other in comparison to full price matching such that players undercut each other more often both at the monopoly price level and at the restoration provocation level. The provocation level is again higher compared to without price matching. The consulted firm plays a sharp price decrease anticipating that the competitor's is likely to restore. It is uncertain who leads the price increase in the classic version of the Edgeworth model, while with price matching the implementing firm leads price

Figure 3: Edgeworth Cycle Equilibrium with Partial Price Matching ($\alpha = 0.30$)



The left graph represents the best responses strategy space for both players conditional on the action of the other player. The right graph draws a simulated price path of 30 periods within the infinite game.

restorations. The average posted price of the price-matching firm is on average higher with price matching. The consulted firm may charge lower prices to provoke more price restorations.

To draw conclusions regarding price and welfare implications, I simulate 10,000 moves of the game and compute average prices for different levels of the redemption rate α and list the them in table 2. With price matching, posted prices of the policy-implementing firm increase, while they are lower for the consulted firm. The minimum price of the entire game increases for both firms because they do not undercut each other to marginal costs with price matching. The column *Effective Price: Transactions* gives the average price weighted by transaction volume. Consumers pay a lower average transaction price with a higher share of consumers claiming price matching.[5]. In contrast to static models, the dynamic component drives consumers' being better off with price matching. These welfare improvements increase with higher redemption rates.

The column denoted as *Effective Price: Periods* represents the mean price over all periods. The results indicate that simple period averages do not reflect the welfare implications from a transaction weighted average. It is non-monotonically dependent on the redemption rate.

Based on the simulations, the model determines three main hypotheses from price matching in Edgeworth cycling markets:

Hypothesis 1 ⟨Posted Prices⟩

Price-matching implementing firms post higher prices.

[5]This result is stable of all linear demand specifications with different parameters.

Table 2: Average Prices Drawn from Price Path Simulations

	Posted Price		Minimum Price		Effective Price	
	Firm 1	Firm 2	Firm 1	Firm 2	Periods	Transactions
Without Price Matching	0.72	0.72	0.00	0.00	0.72	0.56
Price Matching ($\alpha = 0.3$)	0.94	0.58	0.30	0.25	0.76	0.51
Price Matching ($\alpha = 0.5$)	0.95	0.56	0.25	0.25	0.75	0.48
Price Matching ($\alpha = 0.8$)	0.87	0.50	0.35	0.30	0.69	0.45
Full Price Matching ($\alpha = 1$)	0.90	0.44	0.30	0.30	0.67	0.40

The table lists price predictions from different Edgeworth Cycle models. 10,000 moves in the game are simulated. Posted prices represent prices determined by firms' price decisions. Minimum prices represent the lowest price observed in the entire simulation. Effective price represents the price that is paid by the consumer, either because they shop at the lower priced firm or because they claim a price match. Numbers represented periods means for all columns excluding column *Effective Price: Transactions*, which is weighted by transaction volumes.

Hypothesis 2 ⟨Posted Price Magnitude⟩

Price-matching implementing firms increase posted prices stronger than price-consulted firms.

Hypothesis 3 ⟨Posted Minimum Prices⟩

Price-matching implementing and consulted firms restore prices from a higher price level.

3 Empirical Analysis

The panel consists of 14,845 gasoline stations in Germany and 31,465,877 price adjustments between June 8, 2014 and June 23, 2015. Locations of all gasoline stations are visualized on a map in figure 8 in the appendix. The data is collected and published by the German Market Transparency Unit for Fuels, which belongs to the Federal Institute for Competition. Petrol stations selling more than 1 million liters per year are obliged to report price changes of unleaded gasoline (with and without 10 % ethanol) and diesel within five minutes to the Market Transparency Unit for Fuels. Private information service providers can request access to their database and publish the price information to the public. Gasoline stations do not receive direct access but access the data through information providers. Since June 8, 2014, a service provider publishes all price adjustments. Combined with an additional data set that contains opening hours and that was provided by email by the service provider, the final data set includes information on prices, time of price change, retailer brand, address, geographical coordinates, and opening hours. Within the data set, 255 stations have stopped reporting price adjustments, most likely because they shut down their business. These stations represent less than 2 % of the entire sample size.

The reader might question whether a simple model with two players is able to explain behavior in a complex oligopolistic market. The literature has shown that Edgeworth Cycles are theoretically robust in many complex environments like finite time horizons (Wallner et al., 1999), firm size heterogeneity (Eckert, 2003), capacity constraints and fluctuating marginal costs (Noel, 2008), and spatial differentiation (Wills-Johnson and Bloch, 2010). Eibelshäuser and Wilhelm (2019) show that the Edgeworth Cycle model with two players is suited to explain the developments in the analyzed market over multiple years. Therefore, the author is confident that the presented model identifies the same qualitative hypotheses as a more complex one that extends the Edgeworth Cycle model by price matching.

3.1 Price Matching Policies

On April 1, 2015, HEM, a gasoline retailer with around 4 % market share, implemented price matching at its 349 stations. Providing evidence with a mobile price comparison application that competitors post lower prices, customers can claim the lowest price of all competitors within a 5 km radius around the respective station.

On May 27, 2015, Shell, a gasoline retailer with the second highest market share in Germany, introduced another price-matching policy. For all customers identifying with their loyalty card, Shell matches the lowest price of the ten closest gasoline stations automatically, but adds a markup of 2 Cents per liter (cpl). Regular customers pay the posted price. Shell consider prices of a restricted set of competitors. A mathematical representation of the effective price $p_i^{\text{effective}}$ members pay (in cpl) is,

$$p_i^{\text{effective}} = \min\left(p_i^{\text{posted}}, \min\left(\{p_j^{\text{posted}}\}_{j=1}^{N=10}\right) + 2\right),$$

where i represents a participating Shell station, and j denotes the ten closest competitors of station i.

Some stations that are located on a highway, match the prices of the two geographically previous and following stations on the same highway, only. Shell excludes 102 of their 1,861 stations from price matching, claiming that these stations have incompatible payment systems. Information on all excluded stations is available[6] and incorporated in the analysis.

Both policies incur comparably low hassle costs to the examples mentioned in the literature,

[6] Shell provides the list those stations that do not implement price matching and of those that prices are consulted at `https://www.shellsmart.com/smart/promotion?pId=1&site=de-de`. Accessed on May 16, 2019.

Table 3: Descriptive Statistics

Variable	Mean	Std. Dev.	N
Average daily price diesel	1.281	0.088	5,364,096
Average daily price E10	1.446	0.097	5,126,789
Average daily price E5	1.477	0.101	5,160,496
Price adjustments per day (E5)	6.166	3.100	5,160,496
Stations consulted by HEM's price matching	18.6 %	0.389	14,499
Stations consulted by Shell's price matching	67.6 %	0.468	12,984

Prices denoted in Euro. The number of prices differs across products because some retailers do not sell all types of gasoline.

as in Hviid and Shaffer (1999), where customers have to discuss price matches with retailers' representatives and where uncertainty plays a role because of imprecise terms and conditions. For Shell, the hassle cost consists of the 2 cpl markup and the cost of registration with the loyalty program. For HEM, hassle costs represent the cost of using the mobile application that provides information about the exact price match.

Business client cards are excluded from claiming price matches by both firms. This indicates that the focus of gasoline retailers are private customers that use the policy and that posted prices are the relevant variable.

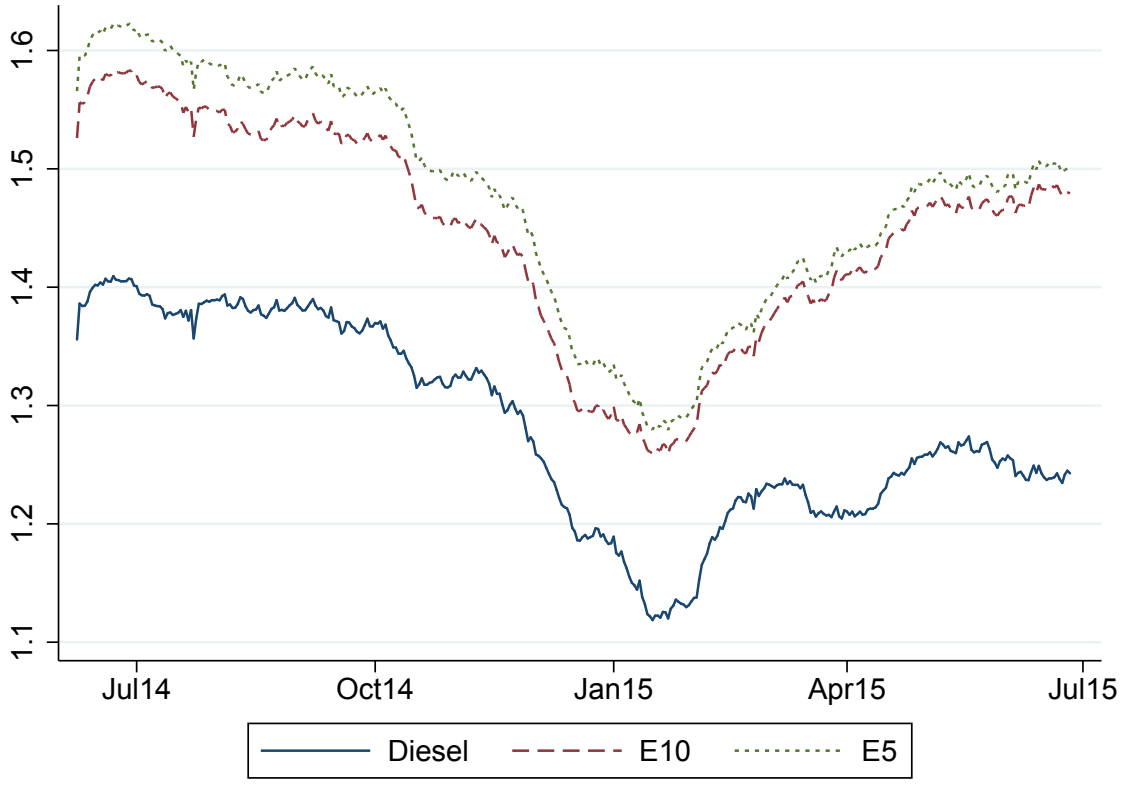
3.2 Descriptive Statistics

Similar to Haucap et al. (2015), reported prices $\geq$ 2 Euro and $<$ 0.90 Euro are regarded as implausible and, thus, excluded from the analysis. This reduces the number of observations by 0.01 %. Average prices are weighted by the time they were active during opening hours. figure 4 plots the average price $\bar{p}_t$ for diesel, 95 RON unleaded gasoline (E5) and 95 RON with up to 10 % ethanol unleaded gasoline (E10).

3.3 Price Strategies

Table 3 summarizes average retail prices. Diesel is the lowest priced type of gasoline because of a lower tax rate. The price of E10 is on average 3 Cents per liter (cpl) lower than E5. Around 5.5 % of the gasoline stations do not sell E10.

Figure 4: Price over Time by Gasoline Type



The figure plots average prices of diesel, E5, and E10 against time. The currency of the price is Euro. The graph reports prices from June 8, 2014, until June 26, 2015.

Table 4 compares the price strategies of price-matching implemented retailers with consulted stations. Over the entire time period, HEM charged on average 2 cpl less than geographically close competitors[7]. Shell's stations may be still more attractive and face higher station provision costs, but it is noted that Shell would rather follow a quality-leadership strategy with high retail prices compared to HEM which follows a low-price strategy. The author is not aware of other structural price strategy adjustments by both price-matching retailers besides the implementation of this policy.

Eibelshäuser and Wilhelm (2019) show that Edgeworth Cycles are able to explain the daily price pattern observed in the analyzed market. At the time of the price-matching implementation, they observe one daily price cycle visualized in figure 5. While 60 % of stations are closing at night, the remaining retailers post the highest prices. After all competitors start operating in

[7]Geographically close competitors are defined following the definition of the gasoline retailers' terms and conditions for price matching: HEM consults every gasoline station that is closer than 5 km (linear distance). Shell defines the 10 closest gasoline retailer as relevant competitors.

Table 4: Price Differences with Price Matching

Variables	Obs. Introd.	Mean Introduced	Obs. Cons.	Mean Consulted	Difference
E5 Price HEM (before introduction)	99,013	1.452	746,492	1.473	-0.021***
E5 Price HEM (after introduction)	30,833	1.455	228,497	1.476	-0.021***
E5 Price Shell (before introduction)	640,316	1.501	2,758,411	1.472	0.029***
E5 Price Shell (after introduction)	61,199	1.532	268,799	1.489	0.042***

This table reports the price differences of the policy implementing retailers and their self-defined close competitors on E5. Consulted stations are geographically close to a policy-implementing station and, therefore, referenced for price matching. All prices are reported in Euro. Standard errors are clustered at the station level. *** p<0.01, ** p<0.05, * p<0.1

the morning, retailers undercut each other until the evening. For Shell, I find a slight price increase that is driven by a price increase experiment half a year before the introduction of price matching, and that was ceased in early 2015.

3.4 Specification

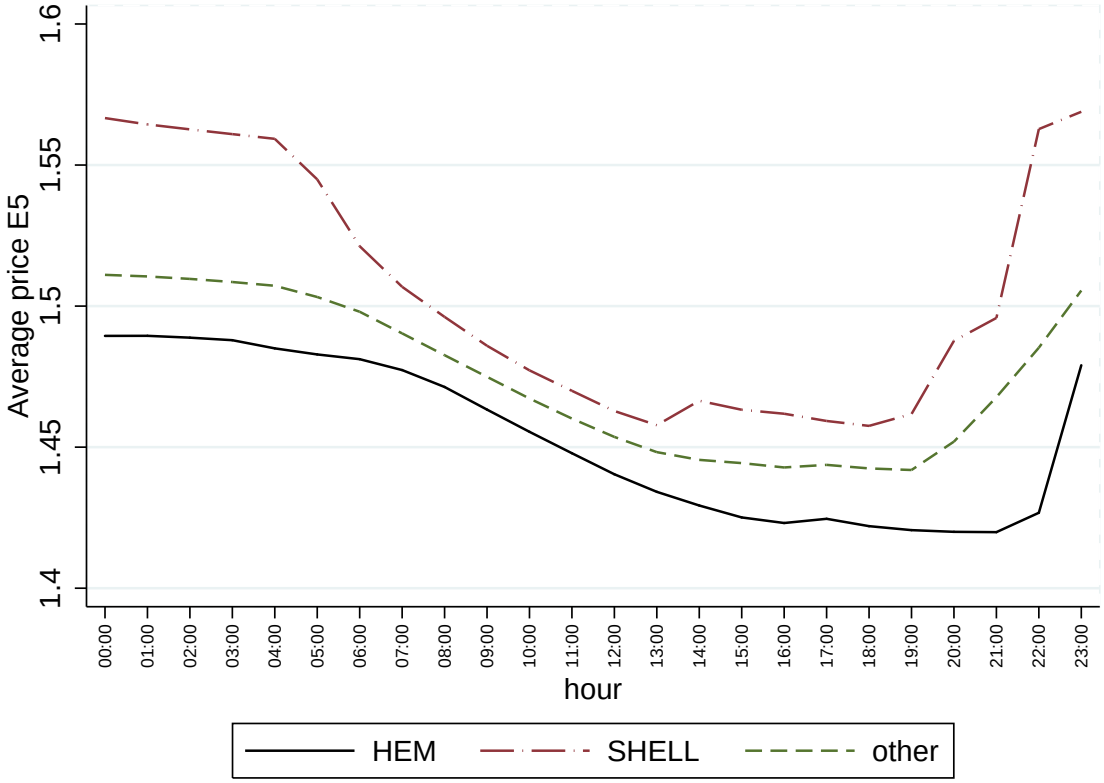
The econometric specification of the model to explain the posted price $\bar{p}_{it}$ is given by

$$\bar{p}_{it} = \gamma_i + \delta_t + \text{BESTPRICE}_{it}\beta + \epsilon_{it},$$

$$\text{BESTPRICE}_{it} = \begin{pmatrix} \mathbb{1}_{\text{HEM brand} \wedge \text{Price matching}} \\ \mathbb{1}_{\text{Station is consulted by HEM price matching}} \\ \mathbb{1}_{\text{Shell brand} \wedge \text{Price matching}} \\ \mathbb{1}_{\text{Station is consulted by Shell price matching}} \\ \mathbb{1}_{\text{Station is excluded by Shell's policy}} \end{pmatrix}$$

where $\mathbb{1}$ represents the indicator function. i represents a station such that γ_i absorbs station fixed effect and δ_t denotes a day-dummy with t representing a day index. ϵ_{it} is the error term. A station is defined as a location where gasoline is sold. A retailer may sell gasoline at more than one station. BESTPRICE_{it} is a vector that represents which kind of price matching the station is affected by. The first entry of the vector is 1 if the station is a price-matching HEM retailer, the second dummy identifies stations that are consulted by HEM's price-matching

Figure 5: Daily Prices Patterns by Type of Gasoline



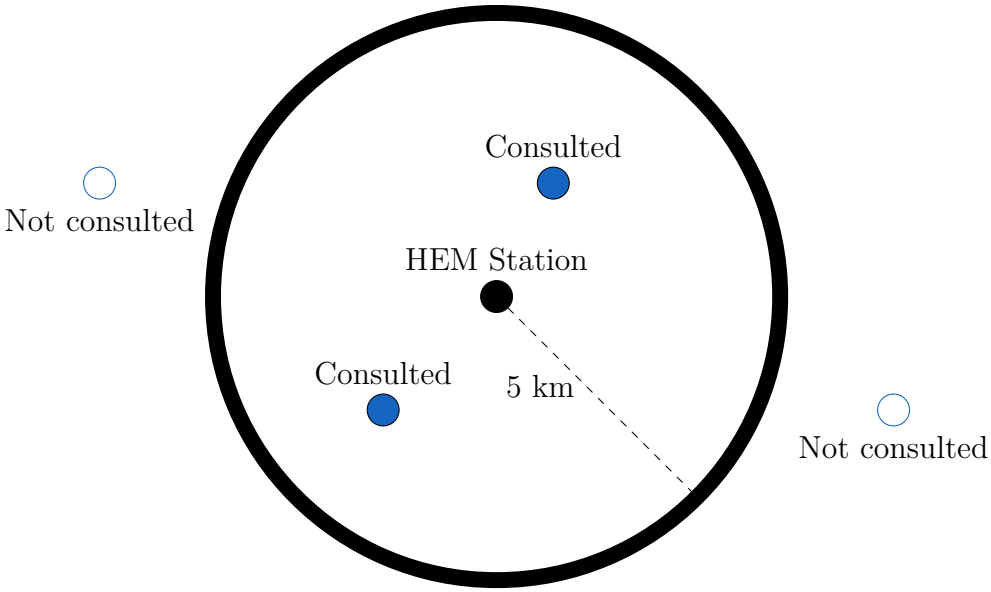
The table reports the average price of E5 type of gasoline for HEM, Shell, and other gasoline retailers in Germany depending on the daytime. Prices are denoted in Euro.

policy. The third captures effects for Shell stations that implemented price matching, and the fourth identifies stations that are consulted by Shell's price-matching policy. The last interaction term indicates stations being one of the closest 10 to Shell stations but that are excluded by Shell's terms and conditions. The coefficient of the last dummy is not interpreted but shown in all regressions to rule out unanticipated interactions.

The vector β consists of 5 coefficients capturing the price-matching effects for implementing and consulted stations. The different dummies do not exclude each other. Up to two dummies are possibly active by construction. Interactions between different treatments turned out to be economically insignificant and are, therefore, omitted.

Figure 6 and 7 visualize the price-matching policies that determine the value of the interaction dummies. HEM's implementing dummy identifies every observation representing HEM stations after April 1, 2015. HEM's consulted dummy identifies every observation after April 1, 2015, that belongs to a gasoline station which is within a radius of 5 km around a HEM station and,

Figure 6: Sketch of HEM's Price Matching Policy



Consumers of HEM can match the lowest price of all stations around 5km of the station in which they claim the price. All HEM stations are identified as implementing stations, all stations within the 5km radius as consulted, and all other stations as unaffected.

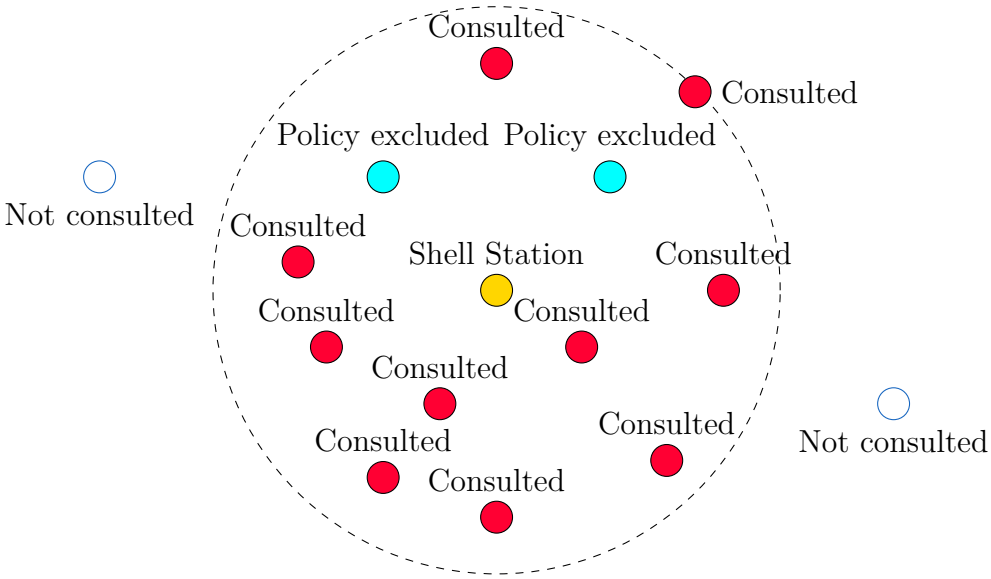
thus, references for price matching. Shell's implementing dummy indicates all Shell stations participating in price matching. Shell's consulted dummy identifies stations that are one of the ten closest participating stations of Shell, and which are not excluded by Shell's terms and conditions. The control group consists of all observations of stations that are not affected by price matching, because either they are not consulted or because the observation relates to an observation before any price-matching policy is implemented.

3.5 Test of Hypothesis 1: Posted Prices

To test whether price matching implementing firms set higher posted prices, I regress prices of different gasoline types on the treatments controlling for day and station fixed effects. Table 5 lists the estimated coefficients.

The evidence is divided. Shell's coefficient on implementation is highly statistically significant and accounts for around 1.5 cpl price increase for all types of gasoline. For HEM, the coefficient is also positive for diesel but negative for E10 and E5. In comparison to the estimated effects on Shell's prices, HEM's coefficients are rather low. Hence, while Shell's behavior is in line with hypothesis 1, HEM's pricing might not be reacting because of the relatively small size of HEM and its limited impact on the market.

Figure 7: Sketch of Shell's Price Matching Policy



Consumers of Shell can match the lowest price with a mark-up of 2 cpl of the ten closest stations. Some Shell stations do not participate. Some non-branded competitors are excluded in Shell's terms and conditions. All participating Shell stations are identified as implementing stations, the ten closest and not excluded competitors as consulted, all closest and excluded competitors as excluded, and all other stations as unaffected.

3.6 Test of Hypothesis 2: Posted Price Magnitude

To test whether implementing stations increase posted prices stronger than consulted competitors, I use a one sided F-test on the estimated coefficients in table 5 with the purpose to falsify the following null hypothesis

$$\beta_i^{\text{Implementing}} - \beta_i^{\text{Consulted}} \leq 0, \; i \in \{\text{Shell, HEM}\}.$$

When the hypothesis is rejected, we can expect that the implementing price matching firm posts higher prices with price matching. As the hypothesis is rejected (p-value < 0.0001), the model is verified for Shell's price-matching policy. For HEM, however, the hypothesis is only rejected for the diesel type of gasoline (p-value < 0.0001). For HEM's E10 and E5 type of gasoline, the hypothesis is not rejected (p-value > 0.9) and possibly larger than zero.

3.7 Test of Hypothesis 3: Posted Minimum Prices

I analyze station's daily minimum prices to test whether implementing and consulted stations increase posted prices with price matching. Table 6 presents the estimated coefficients for a

Table 5: Fixed Effects Regressions on Day-Average Prices

Variables	(1) Diesel	(2) E10	(3) E5
HEM Implementing	0.00250***	-0.00144***	-0.00154***
	(0.000497)	(0.000470)	(0.000467)
HEM Consulted	-0.00152***	-0.000769***	-0.000683**
	(0.000293)	(0.000297)	(0.000289)
Shell Implementing	0.0151***	0.0161***	0.0158***
	(0.000367)	(0.000358)	(0.000352)
Shell Consulted	-4.21e-05	0.00106***	0.00113***
	(0.000341)	(0.000325)	(0.000318)
Shell Excluded	0.00221***	0.00194***	0.00150***
	(0.000382)	(0.000475)	(0.000375)
Observations	5,323,106	5,087,338	5,121,024
Adjusted R^2	0.965	0.971	0.975
Station Fixed Effects	Yes	Yes	Yes
Day Fixed Effects	Yes	Yes	Yes

Robust standard errors clustered on the station level in parentheses. * p<0.10, ** p<0.05, *** p<0.01

fixed effects regression of minimum prices on the treatment dummies for all types of gasoline.

The estimates provide evidence for the model as Shell-implementing and Shell-consulted stations set higher minimum prices after the introduction of price matching. For HEM's implementing stations, however, the negative coefficient estimates contradict the hypothesis. Coefficients on HEM's consulted competitors have both positive and negative coefficient signs depending on the type of gasoline and remain inconclusive.

4 Comparison with Alternative Models

The hypothesis testing section determines mixed evidence for the importance of the developed dynamic model. While the behavior of Shell is in line with all hypotheses, the pricing behavior of HEM only partially comply with the model. The author can speculate about the reasons. Potentially, the crucial assumption of the Edgeworth Cycle model that firms price decisions have a substantial importance on other firms' payoff is not fulfilled because HEM's retailer network is too small and might not be able to shift competitors' profits sufficiently. To find

Table 6: Fixed Effects Regressions on Day-Minimum Prices

Variables	(1) Diesel	(2) E10	(3) E5
HEM Implementing	-0.00245***	-0.00121**	-0.00127**
	(0.000594)	(0.000538)	(0.000536)
HEM Consulted	-0.00138***	0.000156	0.000236
	(0.000239)	(0.000236)	(0.000227)
Shell Implementing	0.00360***	0.00619***	0.00600***
	(0.000313)	(0.000324)	(0.000316)
Shell Consulted	0.00123***	0.00174***	0.00185***
	(0.000235)	(0.000237)	(0.000228)
Shell Excluded	0.000667**	0.00178***	0.00124***
	(0.000304)	(0.000433)	(0.000302)
Observations	5,275,108	5,053,038	5,086,661
Adjusted R^2	0.954	0.959	0.965
Station Fixed Effects	Yes	Yes	Yes
Day Fixed Effects	Yes	Yes	Yes

Robust standard errors clustered on the station level in parentheses. * p<0.10, ** p<0.05, *** p<0.01

out whether other models are able to fill this gap, this section tests competing models with the empirical results.

4.1 Competition Reducing Motives

Observed price decreases of HEM's implementing and consulted stations contradict economic literature modeling price matching as a competition-reducing or collusion-supporting device. Such models predict that the policy always leads to higher prices for implementing and consulted stations.

Furthermore, the empirical results do not indicate that retailers achieve better price coordination with price matching. By observing a decreasing level of price dispersion, Hess and Gerstner (1991) conclude that price matching supports price coordination. However, I find opposite effects from both policies. Shell, the relatively higher-priced retailer, increases its posted prices more than its lower-priced competitors. For HEM, a relatively lower-priced retailer, the opposite is the case. Hence, price matching boosts price dispersion in both policies.

4.2 Signaling Motives

Models explaining price matching as a low-price signaling device predict that implementing low-priced firms increase their prices (Moorthy and Zhang, 2006), while HEM, a retailer with a low retail price strategy, decreases its prices. Additionally, consulted firms decrease their prices with price matching in signaling models. However, I determine a price effect that coincides with the sign of the effect from the policy-implementing company. Furthermore, in the model by Moorthy and Zhang (2006), other inexpensive retailers would also have to implement price matching in equilibrium. In contrast to the model's prediction, until 2019, no other retailer has implemented a similar policy after Shell's and HEM's implementation in 2015.

4.3 Price Discrimination Motives

Corts (1997) shows that price matching may be useful to discriminate in prices between informed and uninformed consumers. In contrast to other models, his model is, generally, able to explain increases and reductions of posted prices.

For low transportation costs, and, thus, a high elasticity of demand for sophisticated customers, the model predicts price increases; whereas for high transportation costs, it predicts decreasing prices. Shell's and HEM's stations are distributed across the entire German retail gasoline market and it is hard to argue that customers face very heterogeneous transportation costs across retailers such that price effects are as dispersed as estimated.

While HEM follows a leadership strategy in the market and charges relatively low prices at its stations, Shell rather follows a quality leadership pricing strategy as shown in table 4. Shell's policy, moreover, charges a markup of two cpl with price matching, indicating that it is reasonable to expect a higher elasticity of demand for HEM's customers. The estimates, however, are exactly the other way around than predicted by the price discrimination model. Shell's low demand elasticity stations increase and HEM's high demand elasticity stations decrease prices significantly, contradicting a main hypothesis of Corts (1997).

5 Conclusion

Integrating price matching into the dynamic Edgeworth Cycle model has strong implications on the price cycle pattern. Posted prices of implementing firms increase and price restorations

occur more frequently. The price level to which consulted firms undercut implementing retailers is at a higher level than in the original model. These predictions indicate that price matching may have competition-reducing implications in cycling markets. Customers, however, are not necessarily suffering as they obtain the opportunity to shop at a low price more frequently. My model shows that the effects from price matching enhances consumer welfare with a higher redemption rate leading to more benefits.

Eibelshäuser and Wilhelm (2019) show that retailers implement an additional price restoration right after the end of the analyzed data set. The following period is omitted in this article because it introduces a structural change in the behavior of the market participants, which could have heterogeneous effects on different players and regions of the market that cannot be controlled for. They show that the additional price increase is driven by a general trend in the market with faster price reactions by all retailers who process competitors' price information more efficiently. The results from the model show that price matching can accelerate price cycles and could have been a driver for an earlier implementation of the additional price restoration.

Shell's pricing behavior verifies the predictions of higher posted average and minimum prices. For HEM, I find relatively small effects that do not comply with the predicted behavior. Possibly, HEM's market share is too small to affect other retailers with its policy. When the author examined an interview with HEM's management, representative mentioned that they do not regard price matching as a strategic device but rather treat it as a marketing tool to substantiate their image as a low cost retailer. Further research is welcomed to investigate the underlying mechanism of HEM's policy as competing models are also not able to rationalize the observed behavior.

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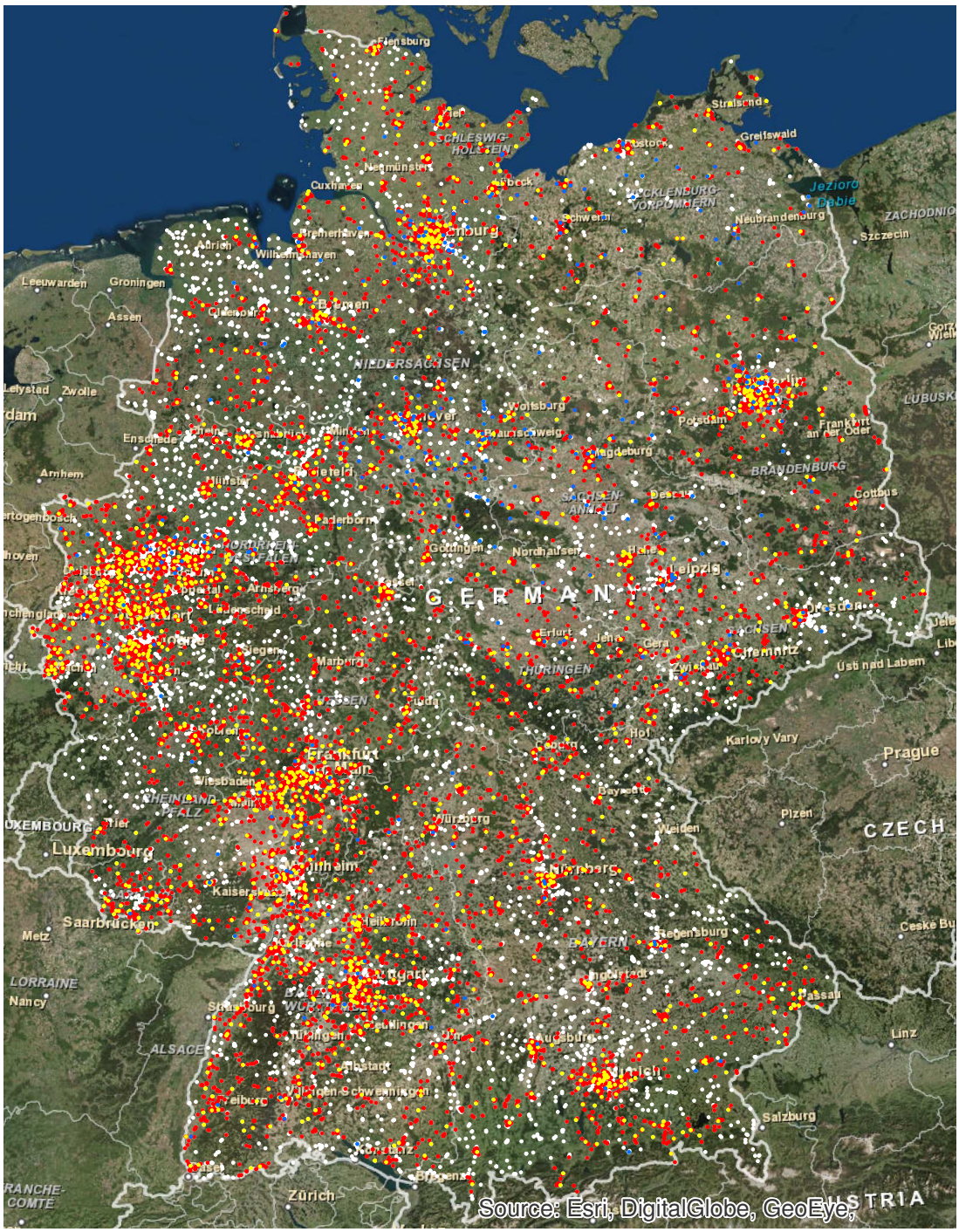
A Appendix

Figure 8: Locations of Observed Gasoline Stations



Blue represents stations of HEM, yellow dots are Shell stations, and red dots are stations consulted by Shell. White represents gasoline stations, which are not affected.